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# COMPUTER NETWORKS

| Ex.No.3 | **Experiments on CISCO PACKET TRACER (Simulation Tool):** |
|---|---|
| Date: | **a)** Tounderstand the environmentofCISCOPACKETTRACERtodesign a simple network.<br>**b)** Analyse the behaviourof networkdevicesusing CISCOPACKETTRACER simulator.<br>Design a simple network with multiple nodes and connect via networking devices available in the library. Perform simulation and trace communication behaviour of specified network devices.<br>1: Use only HUB to design a small network having 4 to 6 hosts<br>2: Use only a Switch to design a small network with 4 to 6 hosts.<br>3: Use both the device (HUB and SWITCH) for a network and find out the functioning difference between switch and hub.<br>**Find out the network topology implemented in your college and draw and label that topology in your observation book.** |

**AIM:** To study the Packet tracer tool Installation and analyse the behaviour of network devices using **CISCO PACKET TRACER**

**INSTALLING PACKET TRACER**

**Windows**

Visit the Cisco Networking Academy website (www.netacad.com).
Log in using your Cisco NetAcad account.
Download the Packet Tracer installer for Windows.
Run the setup file, accept the license agreement, and complete the installation.

**Linux**

Download the appropriate Packet Tracer package for your Linux distribution.

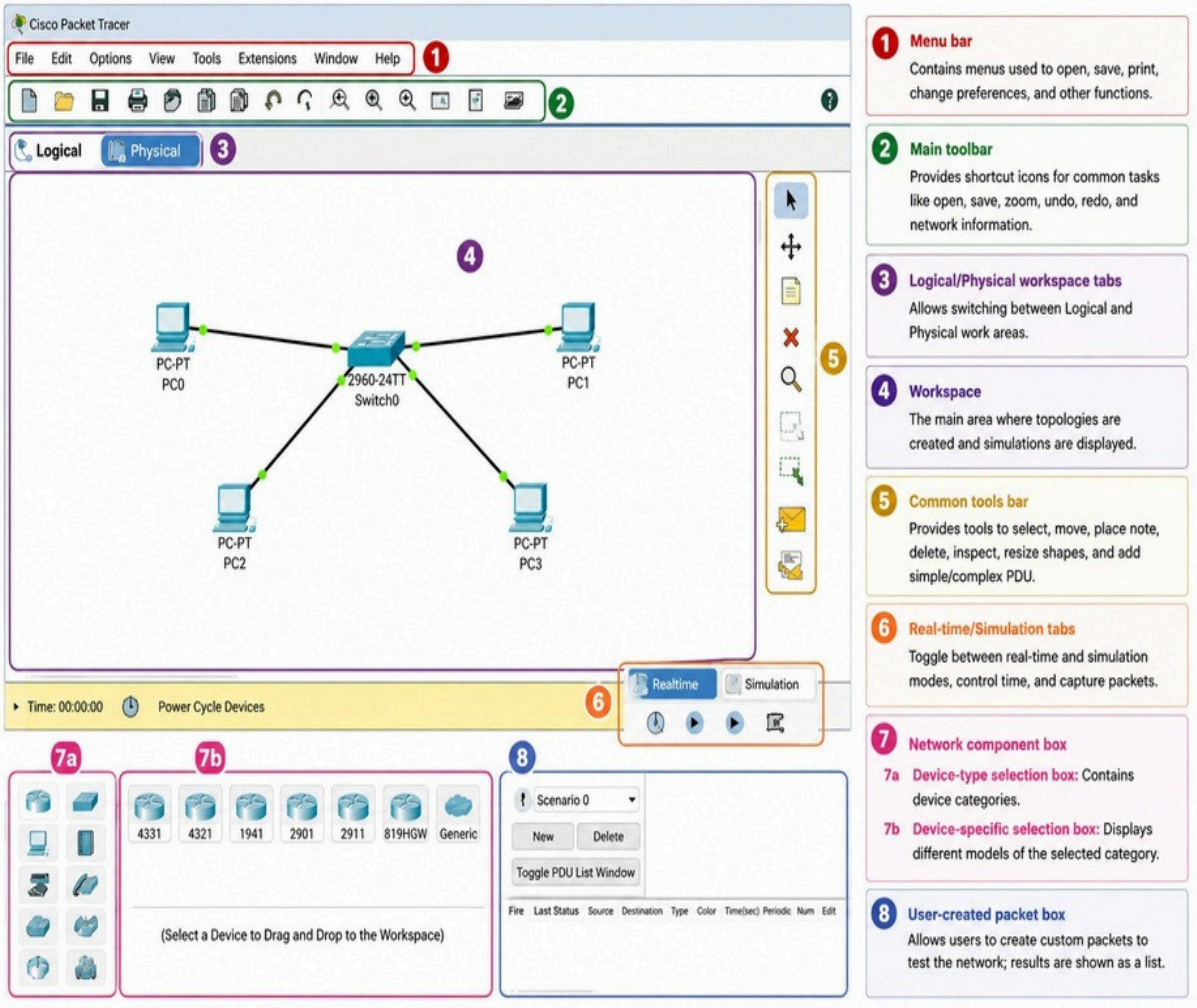
Open the terminal and make the installer executable:

**chmod +x PacketTracer601_i386_installer-rpm.bin**

Run the installer:

**./PacketTracer601_i386_installer-rpm.bin**

Follow the on-screen instructions to complete the installation.




**b) Analyseth e behaviou r of netw ork d ev ices using CISCO PACKET TRACER simulator.**

## Scenario 1: Network Using Only a Hub

**Procedure:**

Step 1

Open **Cisco Packet Tracer**.

Step 2

From **Network Devices**, drag a **Hub** to the workspace.

Step 3
Drag **4 PCs** (PC0, PC1, PC2, and PC3) from **End Devices**.

Step 4
Connect each PC to the Hub using **Copper Straight-Through** cables.

Step 5
Assign the following IP addresses.

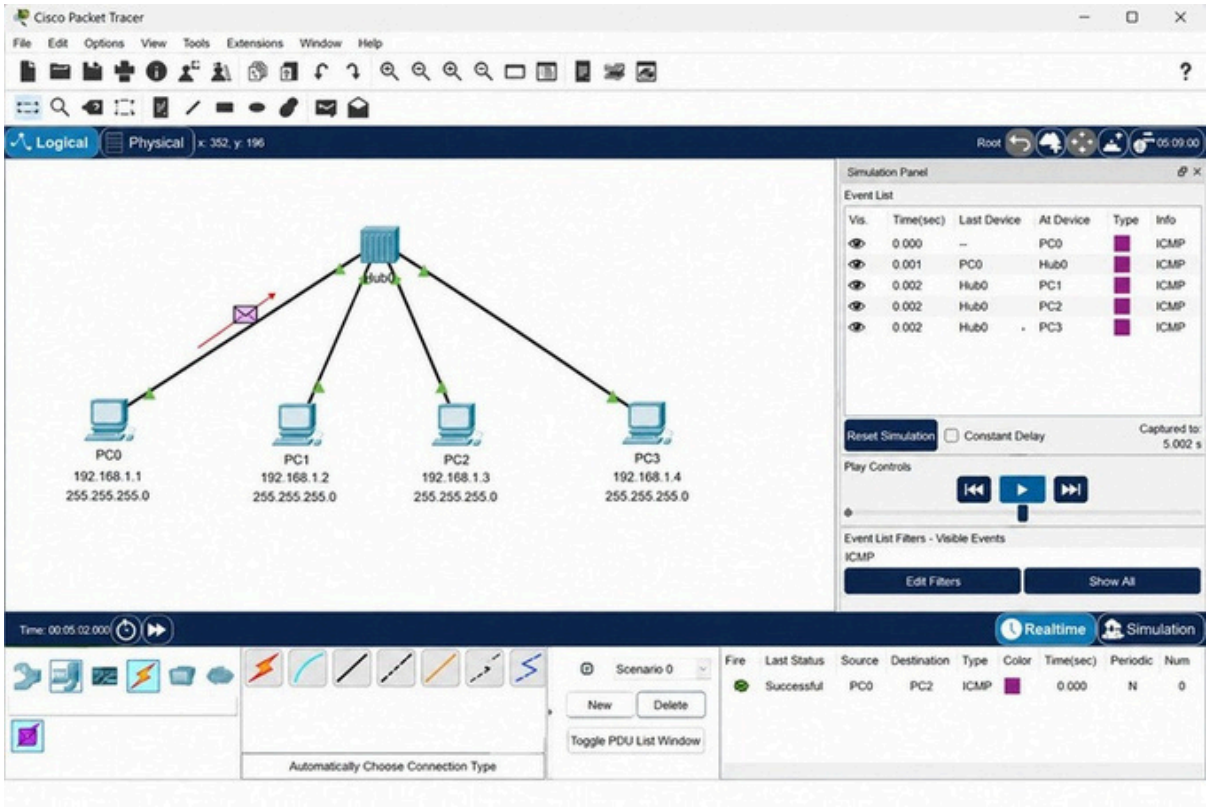
| Device | IP Address | Subnet Mask |
|---|---|---|
| PC0 | 192.168.1.1 | 255.255.255.0 |
| PC1 | 192.168.1.2 | 255.255.255.0 |
| PC2 | 192.168.1.3 | 255.255.255.0 |
| PC3 | 192.168.1.4 | 255.255.255.0 |

Step 6
Switch to **Simulation Mode**.

Step 7
Click **Add Simple PDU** and send a packet from
**PC0** to **PC2**.

Step 8
Observe the packet movement.
**Observation**

- The Hub broadcasts the packet to **all connected hosts**.
- Every PC receives the packet.
- Only the destination PC accepts the packet.
- Network collisions may occur because all devices share the same communication channel.




**c)** Scenario 2: Network Using Only a Switch

**Aim:** To design a simple network using a switch and observe the communication behaviour during simulation.

Network Topology

```
          Switch0
        /   |   |   \
PC0   PC1  PC2   PC3
```

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## Devices Required

| Device | Quantity |
|---|---|
| Switch 2960 | 1 |
| PC-PT | 4 |
| Copper Straight-Through Cable | 4 |

## Procedure

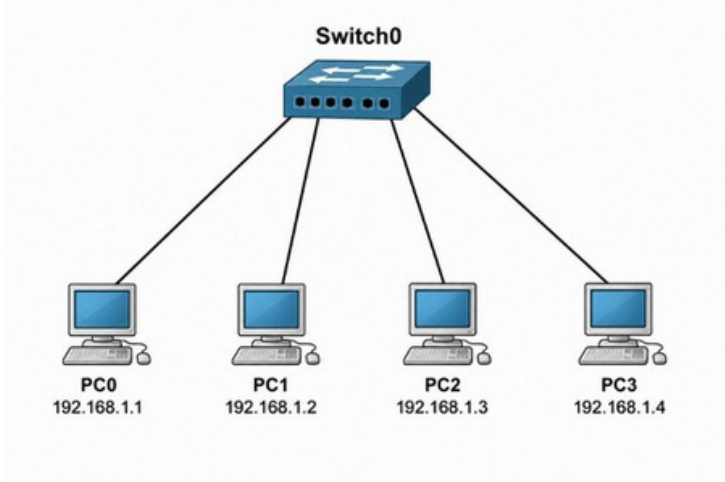
Step 1

Open **Cisco Packet Tracer**.

Step 2
From **Network Devices → Switches**, drag a **2960 Switch** into the workspace.
Step 3
Drag four PCs (PC0, PC1, PC2, and PC3) from **End Devices**.
Step 4
Connect each PC to the switch using **Copper Straight-Through** cables.
Step 5
Assign the following IP addresses.

| Device | IP Address | Subnet Mask |
|---|---|---|
| PC0 | 192.168.1.1 | 255.255.255.0 |
| PC1 | 192.168.1.2 | 255.255.255.0 |
| PC2 | 192.168.1.3 | 255.255.255.0 |
| PC3 | 192.168.1.4 | 255.255.255.0 |

Step 6
Switch to **Simulation Mode**.
Step 7
Select **Add Simple PDU**.
Step 8
Click **PC0** and then **PC2**.
Step 9
Observe the packet movement.




**Observation**

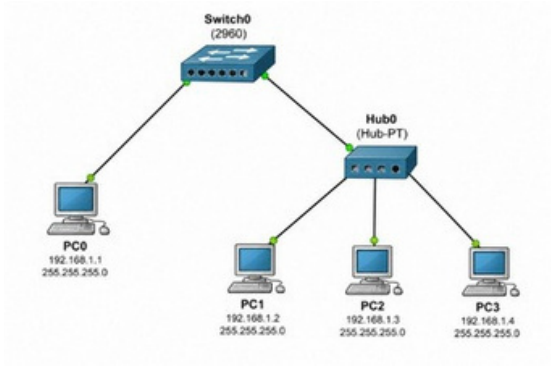
- The switch learns the MAC addresses of connected devices.
- Initially, the switch may flood the frame if the destination MAC address is unknown.

- Once the MAC address table is learned, the switch forwards packets **only to the destination port**.
- Other PCs do not receive the packet.
- No collisions occur because each switch port forms a separate collision domain.

**Scenario 3: Network Using Both Hub and Switch**

**Aim:** To design a network using both a hub and a switch and compare their communication behaviour.

## Network Topology




**Devices Required**

| Device | Quantity |
|---|---|
| Switch 2960 | 1 |
| Hub | 1 |
| PC-PT | 4 |
| Copper Straight-Through Cable | 5 |

**Procedure:**

Open C isco Packet Tracer.

**Step 1** **Step 2** **2960 Switch** and one **Hub** in the workspace.

Place one

**Step 3**

Add four PCs.

PC1 → Hub
PC2 → Hub
PC3 → Hub

**Step 5**
Assign the following IP addresses.

| Device | IP Address | Subnet Mask |
|---|---|---|
| PC0 | 192.168.1.1 | 255.255.255.0 |
| PC1 | 192.168.1.2 | 255.255.255.0 |
| PC2 | 192.168.1.3 | 255.255.255.0 |
| PC3 | 192.168.1.4 | 255.255.255.0 |

**Step 6**
Switch to **Simulation Mode**.

**Step 7**
Send a Simple PDU from **PC2**.

**PC0** to

## Otebps e8r vation

Observe T thhpe swaictkchet fmorowvaermdse ntht.e frame only toward the hub.

- The hub broadcasts the frame to all devices connected to it.
- PC1 and PC3 receive the frame but discard it.
- PC2 accepts the packet because it is the intended destination.
- The switch limits unnecessary traffic, while the hub broadcasts within its segment.

**RESULT:** Therefore the working of the cisco packet tracer and ip configuration of a message to be sent via the internet has been studied, simulated and verified.

